

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

Accurate posterior approximation is a central computational objective in Bayesian inference. Stein Variational Gradient Descent (SVGD) addresses this objective by transporting an interacting particle ensemble toward a target distribution along update directions defined by a reproducing kernel Hilbert space (RKHS). Since the kernel determines this RKHS, it also determines the geometry of particle interaction and thereby exerts direct control over convergence behavior and approximation quality. The widely used median heuristic provides a convenient scalar bandwidth rule, but its limited adaptivity can lead to inaccurate posterior representations, especially in the higher-dimensional settings examined in this paper. We introduce Adaptive SVGD (Ad-SVGd), a kernel-selection procedure for parameterized kernel families that alternates SVGD particle transport with gradient-ascent updates of kernel parameters chosen to maximize the kernelized Stein discrepancy (KSD). The method turns KSD-based convergence principles into an explicit adaptive inference algorithm. In a simplified mean-field analysis, we extend fixed-kernel KSD descent results to adaptive kernel classes and prove convergence of the maximal KSD over the prescribed class. Across the numerical studies considered here, Ad-SVGd consistently improves on standard bandwidth heuristics and more accurately represents posterior uncertainty.

1 Introduction

Stein Variational Gradient Descent (SVGD) [SVGD] is a deterministic particle methodology for approximate Bayesian inference and a prominent alternative to traditional Markov Chain Monte Carlo (MCMC) algorithms. SVGD transports an ensemble of particles along directions induced by the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, while restricting admissible directions to the unit ball of a reproducing kernel Hilbert space (RKHS). This RKHS constraint is structurally decisive rather than merely technical: because the kernel fixes the geometry of particle interaction, kernel specification can substantially alter the behavior of the algorithm [Duncan2019OnTG, Nsken2021SteinVG]. Empirically, SVGD particle systems often underestimate posterior variance [DBLP:conf/iclr/BaEGSSWZ22], a failure mode that has reinforced the view that the method becomes less reliable as the state dimension grows. This paper shows that an important part of this limitation can be addressed within the original SVGD transport mechanism by replacing static bandwidth selection with an adaptive kernel-parameter update. By tuning the kernel through maximization of the kernelized Stein discrepancy (KSD), the proposed method selects update geometries that produce more effective transport in the complex and higher-dimensional examples studied here.

1.1 Related work.

Since its introduction [SVGD], SVGD has become an established methodology for approximate Bayesian inference in machine-learning applications [DBLP:conf/uai/LiuRLP17, DBLP:conf/iclr/Messaoudi2020, DBLP:conf/nips/PuGHLHC17, DBLP:journals/tsp/KassabS22]. Its mathematical behavior is now characterized by a substantial theoretical literature: mean-field convergence has been studied in continuous time [LLN2019, Duncan2019OnTG, NEURIPS2020'16f8e136] and discrete time [NEURIPS20203202111c, pmlr-v162-salim22a], finite-particle convergence rates have been derived under several assumption regimes [balasubramanian2024improvedfiniteparticleconvergence, NEURIPS2023'54e5d7af], and the dynamics have been interpreted as gradient flows on spaces of probability measures [DBLP:conf/nips/Liu17, Duncan2019OnTG], in analogy with Variational Formulation.

Kernel choice is central to both the empirical performance and the theoretical interpretation of SVGD, because it specifies particle interaction and enters directly into the convergence criterion (see also Figures 1 and 9). Existing convergence statements are commonly expressed in terms of the KSD, and whether KSD convergence entails weak convergence depends on the selected kernel [pmlr-v70-gorham17a]. The median heuristic [JMLR:v13:gretton12a] remains attractive because of its simplicity, but it has no general theoretical justification in this setting and its performance can deteriorate as dimensionality increases [DBLP:conf/iclr/BaEGSSWZ22, DBLP:conf/icml/ZhuoLSZCZ18, DBLP:conf/icml/WangZ018]. Several SVGD variants have therefore been proposed to counteract high-dimensional performance degradation [stein'newton, NEURIPS2019'5dcd0ddd, gong2021sliced, pmlr-v151-liu22a], while related approaches learn update directions with neural networks [langosco2022neural, ZHAO2023118975].

The closest precedent is the mixture-of-kernels framework of ai'stein'2023, which defines a KSD for a weighted linear combination of base kernels and learns the weights by maximizing the resulting multiple-kernel KSD. Our work retains this KSD-maximization principle while moving beyond finite kernel bases: it optimizes continuous kernel parameters, including dimension-dependent bandwidths, and thereby treats kernel adaptation as an intrinsic component of SVGD rather than as a finite mixture-selection problem.

1.2 Contributions.

This work asks whether the original SVGD transport mechanism, without additional architectural or algorithmic modifications, can deliver accurate posterior approximations when equipped with a principled adaptive kernel. By isolating kernel selection from the broader family of SVGD enhancements, the paper clarifies the extent to which the kernel alone governs algorithmic effectiveness and demonstrates that adaptive kernel learning can substantially improve the behavior of the original method.

Our main contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We introduce Ad-SVG, which updates kernel parameters during inference by ascending the KSD before applying the SVGD particle update. Unlike the median heuristic, which imposes a scalar bandwidth rule, the method accommodates multiple continuous kernel parameters and therefore supports anisotropic, iteration-dependent update geometries.
- (ii) **Theoretical Analysis.** We analyze the algorithm in the discrete-time mean-field setting and extend fixed-kernel SVGD convergence results to an adaptive kernel class. In particular,

we prove convergence of the supremal KSD over the parameterized class and, under a Stein logarithmic Sobolev inequality, derive mean-field iteration-complexity bounds.

- (iii) **Empirical Validation.** We show on Gaussian mixtures, Gaussian targets of increasing dimension, inverse problems, Gaussian process inference, and Bayesian logistic regression that adaptive kernel selection improves the approximations obtained with the median heuristic and mitigates the variance collapse observed in the baseline.

2 Mathematical background

We fix a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ and denote its associated RKHS by $\mathcal{H}_0$. The vector-valued RKHS used by SVGD is the d -fold Cartesian product $\mathcal{H}$, equipped with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and the canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. Throughout, $x \cdot y$ denotes the Euclidean inner product and $\nabla \cdot$ denotes divergence. We write $\mathcal{P}(\mathbb{R}^d)$ for the space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, and $\mathcal{P}_p(\mathbb{R}^d)$ for the subset with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the Wasserstein p -distance is

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ whose marginals are μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [IPM] quantify the discrepancy between two probability measures by optimizing expectation differences over a prescribed class of test functions. Stein’s method [bounderrorno] adapts this variational principle to intractable targets through operators whose expectations vanish under the target distribution. For the Stein operator $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable kernel, one has $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [KernelTestGoodness, KernelizedSteinDiscrepancy]. The resulting *kernelized Stein discrepancy* (KSD) is defined as

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. The optimizer is $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, where $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$ and Φ_k is the feature map associated with k . Thus the variational problem is represented by the RKHS norm of the Stein witness function, yielding the equivalent characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and an initial distribution μ_0 , SVGD approximates π through the push-forward recursion $\mu_{n+1} := T_{\#} \mu_n$, where $T_{\#} \cdot$ denotes push-forward by $T(x) = x + \gamma \psi^{\mu_n}(x)$ and $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ for $\mu \in \mathcal{P}(\mathbb{R}^d)$ is the RKHS steepest-descent direction. This update follows from Equation (1): $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ minimizes the directional derivative $\frac{d}{d\gamma} \text{KL}(T_{\#} \mu_n \| \pi) \Big|_{\gamma=0}$ over the unit ball

of $\mathcal{H}$ [SVGD], where $\text{KL}(\cdot \| \cdot)$ denotes the KL-divergence. For this direction,

$$\frac{d}{d\gamma} \text{KL}\left(\left(\text{Id} + \gamma \psi^{\mu_n}\right)_\# \mu_n \parallel \pi\right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n | \pi). \quad (2)$$

Applying these transport maps to particles $\{X_0^i\}_{i=1}^M$ sampled from μ_0 gives the finite-particle SVGD recursion

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because KL convergence for SVGD is difficult to establish without restrictive assumptions, recent analyses often formulate convergence in KSD [NEURIPS20203202111c, NEURIPS2023'54e5d7af, pmlr-v162-salim22a]. KSD convergence alone, however, does not automatically imply weak convergence: the corresponding mean-field measures may fail to be tight, which is necessary for weak convergence on Polish spaces. Moreover, Equation (1) makes explicit that the RKHS—and therefore the kernel—determines how informative KSD convergence is about convergence of probability measures [pmlr-v70-gorham17a].

We therefore formulate SVGD over a parameterized kernel family $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$, let KSD_θ denote the corresponding *kernelized Stein discrepancy*, ψ_θ^μ the associated optimal update direction, and Φ_θ the corresponding feature map. This parameterization elevates kernel choice from a preprocessing convention to an optimization variable, allowing the RKHS geometry to adapt to the current particle approximation rather than remaining fixed throughout inference.

The adaptive rule follows a direct variational principle: at the current particle measure, select the kernel that maximizes the KSD to the target. Since the instantaneous KL decrease under SVGD is proportional to the squared KSD (see Equations (2) and (7)), KSD maximization selects, within the prescribed kernel class, the update geometry associated with the largest first-order decrease. This interpretation is consistent with geometric analyses of SVGD as a gradient flow, in which kernels producing larger KSD values yield more favorable convergence behavior when the corresponding RKHSs are compared [Nskn2021SteinVG, Duncan2019OnTG].

Whereas standard SVGD usually studies KSD convergence for a fixed kernel, Ad-SVGd targets the largest KSD attainable within the selected kernel class. In contrast to the median heuristic used by **SVGD**, the mechanism applies to any parameterized kernel family for which the KSD can be evaluated and differentiated. At iteration n , the idealized kernel-selection objective is

$$\theta_n \in \underset{\theta \in \Theta}{\operatorname{argmax}} \text{KSD}_\theta(\mu_n | \pi).$$

The practical algorithm approximates this maximization by performing one or more gradient-ascent steps with step size $s > 0$ on the KSD before the particle update, and it permits kernel updates to be skipped on selected SVGD iterations to control runtime (see Algorithm 1). This option is governed by the user-specified decision variable `paramupdate`. The particle-transport step can be combined with standard SVGD refinements, including adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates, without changing the kernel-selection principle. Likewise, the gradient-ascent update for the kernel parameter can be replaced

by another iterative optimizer. Our implementation differentiates the explicit KSD representation [**KernelTestGoodness**, **KernelizedSteinDiscrepancy**]

$$\text{KSD}^2(\mu|\pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_\pi^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

which permits direct differentiation with respect to the kernel parameters. The additional cost relative to vanilla SVGD is controlled by two implementation features:

- The gradient-ascent steps for the kernel parameter reuse the same score evaluations $\nabla \log \pi(X_n^i)$ as the corresponding SVGD step, so Ad-SVGd requires no additional evaluations of $\nabla \log \pi$.
- The computational burden can be reduced by taking a small number of kernel-parameter steps `nstepstheta`, by updating the kernel only after several particle updates, or, for large particle ensembles, by estimating the KSD from a particle subsample.

Algorithm 1: Ad-SVGd

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps `nsteps`, `nstepstheta` $\in \mathbb{N}$

Output: Final particle set $\{X_{\text{nsteps}}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```

for  $n = 0$  to nsteps  $- 1$  do
    Decide paramupdate  $\in \{\text{True}, \text{False}\}$ ;
    if paramupdate then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to nstepstheta  $- 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{\text{nstepstheta}}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end

```

4 Convergence analysis

We consider a target measure with Lebesgue density of the form $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. The analysis formalizes the adaptive-kernel motivation by extending recent KSD

convergence results for SVGD to a parameterized kernel class. Specifically, we adapt the convergence argument of [pmlr-v162-salim22a], which is developed under Talagrand’s inequality; this condition holds under mild assumptions on the target distribution and is weaker than the logarithmic Sobolev inequality commonly used in related analyses. We impose the same assumptions as [pmlr-v162-salim22a], uniformly over the kernel-parameter space.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. *We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand’s inequality T1, which means that there exists $\lambda > 0$ such that*

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \| \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. *We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.*

4.1 Convergence analysis of SVGD under Talagrand’s inequality

We write

$$\mathcal{F}(\mu) := \text{KL}(\mu \| \pi).$$

and begin from the following fundamental inequality. For a fixed kernel parameter $\theta \in \Theta$ satisfying Assumption 3, define the push-forward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for an arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, let $\gamma, B > 0$, $\alpha > 1$, and $g \in \mathcal{H}_\theta$ satisfy $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Proposition 3.1 of [pmlr-v162-salim22a] then gives

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta}^2 \quad (6)$$

with $K = (\alpha^2 + L)B$. For the iterates (3) with a fixed kernel parameter $\theta \in \Theta$, this inequality yields the descent condition [pmlr-v162-salim22a]

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1}. \quad (8)$$

The essential step in verifying (7) is the use of Talagrand’s inequality, Assumption 2, to establish $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, which permits the application of (6) with $g = \psi_\theta^{\mu_n}$. The descent condition (7) then implies

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

because $\sum_{n=0}^\infty \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$.

For an adaptive kernel-parameter sequence (θ_n) , the same descent mechanism takes the form

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGD.

We first analyze the idealized adaptive SVGD iteration

$$\begin{aligned}\mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi),\end{aligned}\tag{9}$$

so that the selected parameter attains the largest KSD-induced descent direction within the class:

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu_n | \pi).$$

This exact formulation is available when the KSD maximization over kernel parameters has a closed-form solution. The multiple-kernel SVGD framework [ai'stein'2023] provides such a case by parameterizing the kernel as a convex combination of base kernels,

$$k_{\theta}(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

For the general parameterized setting considered by Ad-SVGD, the kernel update θ_n in (9) is typically obtained only approximately. In this general setup we assume that $\operatorname{argmax}_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The following descent condition transfers Theorem 3.2 of [pmlr-v162-salim22a] to the exact adaptive choice and replaces the fixed-kernel KSD by the maximal KSD over the kernel class:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

there exists $c_{\gamma} > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \operatorname{KSD}_{\theta}^2(\mu_n | \pi) = 0$.*

The practically relevant case is approximate maximization of the KSD with respect to the kernel parameter. Suppose that $\max_{\theta \in \Theta} \operatorname{KSD}_{\theta}(\mu | \pi) < \infty$ for any $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned}\theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\operatorname{Id} + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n\end{aligned}\tag{10}$$

for a sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ designed to increase $\operatorname{KSD}_{\theta}(\mu_n | \pi)$. In Algorithm 1, Ψ_n is the gradient-ascent update for the KSD, while the analysis also accommodates other iterative optimization schemes satisfying the same approximation property. The required property of the update rule is the following asymptotic approximation condition.

Assumption 4. We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

This assumption yields the central convergence guarantee for the approximate adaptive scheme: even when the KSD maximization is not solved exactly at each iteration, the maximal KSD over the kernel class still converges to zero.

Theorem 3. Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Theorem 3.2 of [pmlr-v162-salim22a] gives

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and is therefore a Cauchy sequence. Consequently,

$$c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \leq \text{KSD}_{\theta_n}^2(\mu_n | \pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under a Stein logarithmic Sobolev inequality, the adaptive KSD convergence statement strengthens to convergence in KL divergence. For specific regimes of ε_n , we derive iteration-complexity bounds for the mean-field limiting system (10). This refined setting also quantifies the bias incurred when the KSD maximization is solved only to a fixed $\bar{\varepsilon}$ -accuracy at each iteration; the details are given in Appendix B. Importantly, the condition requires only one kernel in the class $\{k_{\theta}\}$ to satisfy the Stein logarithmic Sobolev inequality, preserving substantial flexibility in verifying the assumption.

5 Numerical Experiments

We evaluate Ad-SVGD on a one-dimensional Gaussian mixture model and on a linear inverse problem governed by an ordinary differential equation. Additional experiments in Appendix A examine the effect of increasing dimensionality and Gaussian process inference, while Section 5.4 compares Ad-SVGD with median-heuristic SVGD on the Bayesian logistic regression model used in prior SVGD literature.

5.1 Kernel parameterization

SVGD is most commonly implemented with kernels of the form

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

where $\|\cdot\|_p$ denotes the p -norm on $\mathbb{R}^d$ [SVGD, DBLP:conf/iclr/BaEGSSWZ22, Duncan2019OnTG]. We focus on selection strategies for the parameter h , the kernel *bandwidth*. The standard heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, where *med* is the current median inter-particle distance; this rule is motivated by balancing the two terms in the SVGD update (3) [SVGD].

To exploit the flexibility of the adaptive procedure, we use product kernels of the form

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with parameter $h = (h_1, \dots, h_d)$, thereby allowing dimension-dependent bandwidths. The derivatives required to apply Algorithm 1 with these kernels (see Equations (4) and (5)) are obtained directly from the explicit KSD representation. We also tested an adjusted median heuristic that applies a naive median rule separately in each dimension, but this variant did not produce competitive results and suffered from variance collapse.

Beyond the kernels used in the present experiments, the Ad-SVGD formulation can accommodate more scalable parameterized kernel families. Spectral or low-rank kernel parameterizations, for example, can adapt dominant directions of variability without the full cost of product kernels, while Matérn or mixed-product constructions offer additional control over smoothness and anisotropy.

5.2 Toy example

We first consider the one-dimensional example from SVGD, a two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. We use $M \in \{50, 200, 500\}$ particles, sample the initial particle set from $\mathcal{N}(0, 1)$, and run SVGD for 10^4 steps with step size 1 using the kernels described above with $p = 1$ and fixed bandwidths h . Sample quality is measured by the Wasserstein 1-distance $\mathcal{W}_1$, computed from the explicit formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see StatisticalAspectsWasserstein) and an exact sample of size 10^5 . As shown in Figure 1, accurate approximation occurs only over a restricted bandwidth range, demonstrating that SVGD is highly sensitive to h and that principled bandwidth selection is consequential even in this simple setting.

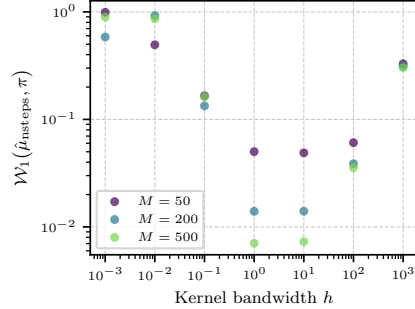


Figure 1: Final Wasserstein 1-distances in the one-dimensional Gaussian-mixture example for SVGD with fixed bandwidths h .

5.3 Linear inverse problem based on ODE

The next experiment, adapted from [Weissmann2022Optimization], evaluates posterior approximation in a linear inverse problem. We consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

and the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete noisy observations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observation model is

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise. The forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is $\mathcal{O} \circ H^{-1}$, with $H(f) = -f'' + f$ and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ denoting observation at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation, we place a Gaussian process (GP) prior on u through the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and $x_k \sim \mathcal{N}(0, \lambda_k)$ independently with $\lambda_k = 50k^{-2}$.

The inferential task is therefore to estimate the KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ under the prior $\mathcal{N}(0, \Gamma_0)$, where $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. With $\varepsilon \sim \mathcal{N}(0, \Gamma)$, the posterior density is

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

For the implementation, we replace H by a numerical discretization operator for Equation (11) using a grid with mesh size 2^{-8} and consider the fully observed system (i.e. $N_{\text{obs}} = 2^8$). We use $N_x = 16$ terms for the KL expansion of u and assume noise covariance $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. We construct reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We compare the performance of SVGD using the median heuristic (we call this Med-SVGd) with our Ad-SVGd for different choices of particle ensemble size $M \in \{50, 100, 200\}$. We use the kernels described in Section 5.1 with the choice $p = 1$. We run both algorithms for $4 \cdot 10^5$ iterations using the step size 10^{-3} for particle updates and, as suggested by **SVGD**, a variant of AdaGrad for adaptive step size control. In Ad-SVGd we use the step size 10^{-5} for the bandwidth updates and update the bandwidth only once for every 100 particle updates. With this update scheme, there was no significant runtime difference between Med-SVGd and Ad-SVGd.

Figure 2 shows the GP reconstruction for an exemplary seed. We observe that both methods are able to give a good approximation of the mean, but only Ad-SVGd correctly captures the posterior

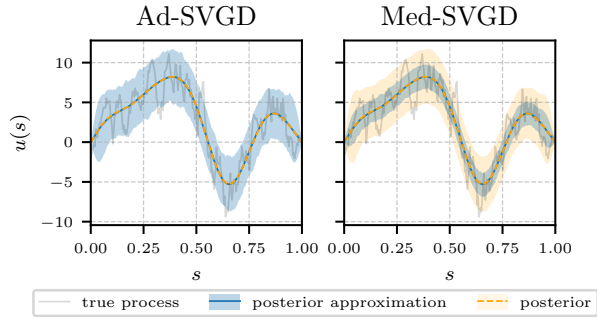
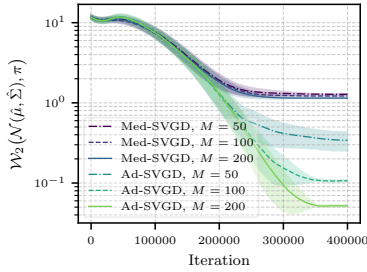


Figure 2: GP reconstruction for the ODE-based inverse problem, with posterior mean and 90% confidence interval.

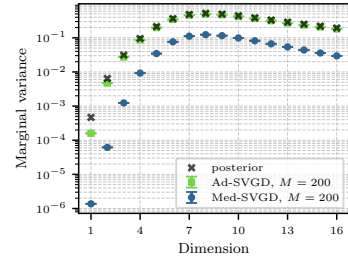
uncertainty. To further quantify the approximation quality, we use the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the normal distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ is the sample mean and $\hat{\Sigma}$ the sample covariance of the particle set [StatisticalAspectsWasserstein]. Since the target π is a multivariate normal distribution, this has an explicit formula

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π are the mean and covariance of the posterior. We also compare the marginal variances of the final particle distribution with the posterior. Figure 3 shows the results of these experiments aggregated over 56 different random seeds (note that the posterior covariance does not actually depend on $\bar{y}$). Again, we observe that Ad-SVGD achieves better approximations of the posterior than Med-SVGD, which underestimates the uncertainty of the problem. Furthermore, in contrast to Med-SVGD, the approximation quality of Ad-SVGD improves as the number of particles increases beyond 50. Figure 4 shows the behavior of the bandwidths determined using Ad-SVGD. We observe that the component-wise bandwidths stabilize more quickly than the approximation error. Clear differences across the components are visible, with the final bandwidths being negatively correlated with the corresponding marginal variances.

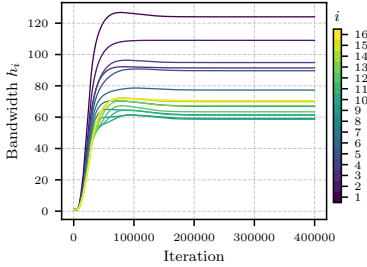


(a) Convergence of approximation error

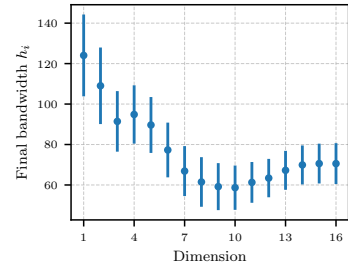


(b) Marginal variances

Figure 3: Aggregated results (mean and 95% confidence interval over 56 random seeds) for ODE-based inverse problem using Med-SVGD and Ad-SVGD



(a) Evolution of bandwidth per dimension (mean)



(b) Final bandwidths (mean and 95% conf. interval)

Figure 4: Behavior of bandwidth parameter for ODE-based inverse problem using Ad-SVGD, aggregated over 56 random seeds

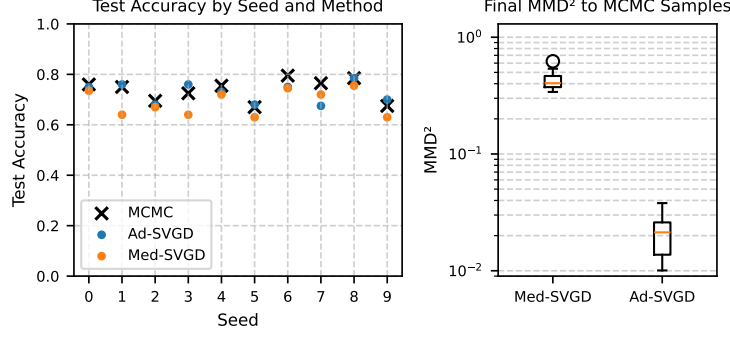


Figure 5: Approximation quality of SVGD particles for Bayesian logistic regression measured by prediction accuracy (left) and MMD^2 to reference samples (right)

5.4 Bayesian logistic regression

As in **SVGD** and **pmlr-v151-liu22a**, we consider a Bayesian logistic regression (BLR) model applied to the Covertypes data set [**covertypes**]. For the regression weights w , we assign a Gaussian prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$ and we want to infer the posterior of $x = [w, \alpha]$. Following **pmlr-v151-liu22a**, we use the No-U-Turn Sampler (NUTS) introduced by **nuts** to generate a reference for evaluating the posterior approximation quality of Med-SVG and Ad-SVG. Again following **pmlr-v151-liu22a**, we test the methods on subsets of the original data set of size 1000 and aggregate our results over 10 random draws. To evaluate the performance of the methods for the classification task, we compute the prediction accuracy over a test set achieved using the particle mean. As shown in Figure 5, Ad-SVG outperforms Med-SVG across almost all seeds, achieving a higher test accuracy while also being closer to the MCMC reference. To measure the accuracy of the posterior approximation, the same figure also shows the squared maximum mean discrepancy (MMD^2) [**MMDref**] to the MCMC samples. We observe a significant difference of how well the two methods are able to capture the posterior distribution. This is further confirmed by Figure 7, which shows the covariance matrices of the final particles compared to the MCMC reference, averaged over the 10 seeds (this visualization style is taken from **pmlr-v151-liu22a**). We observe that Med-SVG severely underestimates the posterior uncertainty, while Ad-SVG gives a better approximation. Additionally, Figure 6 shows the evolution of the bandwidths for each component. We see that from an uninformed initialization (1 in every component), Ad-SVG is able to recover suitable bandwidths, which differ from the initialization by several orders of magnitude.

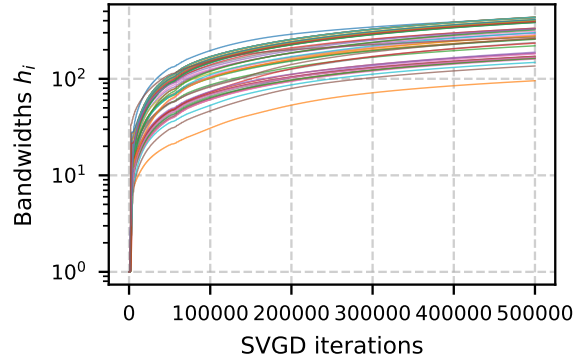


Figure 6: Bandwidth evolution for BLR example

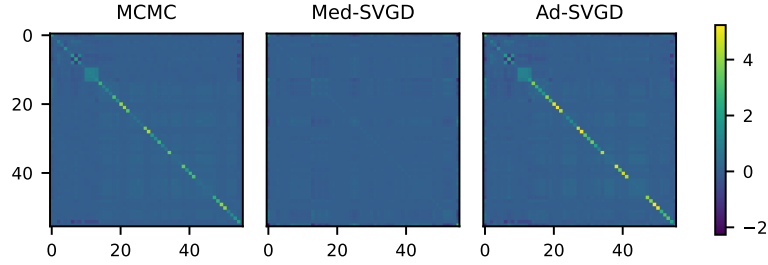


Figure 7: Covariance matrices of final SVGD particle sets compared to MCMC samples (mean over 10 seeds)

6 Limitations and outlook

The main limitation of our analysis is its reliance on Assumption 4. In our ongoing work, we examine when Assumption 4 is satisfied by the alternating gradient-ascent scheme in Algorithm 1 used in our experiments. Although we have no guarantees for Assumption 4 to be satisfied, our implementation led to promising empirical results.

Our considered analysis focused on the original dynamic of the SVGD, and it would be intriguing to combine the proposed adaptive kernel selection with recent variants such as sliced SVGD [gong2021sliced], Grassmann SVGD [pmlr-v151-liu22a], or Stein transport [nsken2024steintransportbayesi]

A More numerical experiments

A.1 Gaussian mixture models

We revisit the example from Section 5.2. For the setup described there, Figure 9 shows histograms of the final particle distribution in comparison to the target density for $h \in \{0.001, 1, 1000\}$. We now turn to the comparison of Med-SVGD with Ad-SVGD. We consider different numbers of particles M and compare the final particle distributions after 10^4 iterations with step size 1 for the two methods.

Figure 8 shows the Wasserstein 1-distance between the final empirical distribution of the particles and the target distribution using Med-SVGD and Ad-SVGD with $M = 10, 20, 50, 100, 200, 500$. We observe that, as expected, the approximation quality improves with N for both methods. Both methods achieve similar results, reaching a Wasserstein distance below 0.01 for $N = 500$.

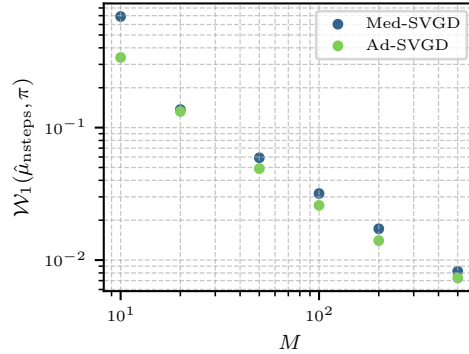


Figure 8: Approximation error for 1D Gaussian mixture.

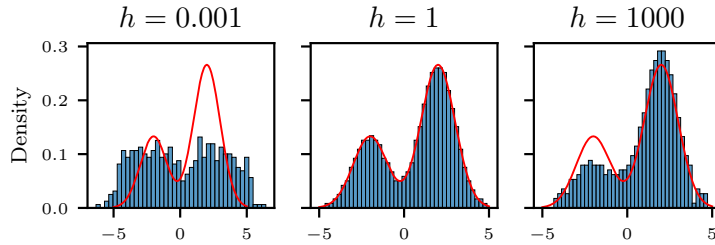


Figure 9: Histograms of final particle set generated using fixed bandwidths $h = 0.001, 1, 1000$; target density π shown in red for comparison.

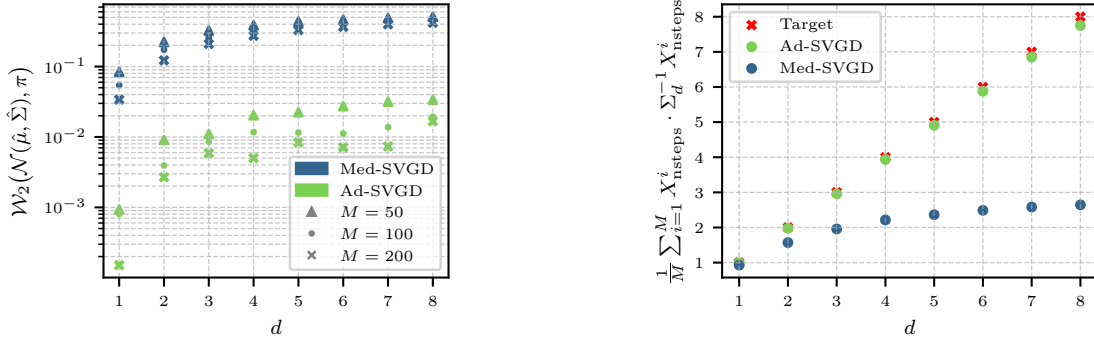
In the one-dimensional case, both methods are able to approximate the target distribution well. Our adaptive bandwidth selection strategy works well, but has no significant advantage over the commonly used median heuristic in this scenario.

A.2 Scaling dimension: multivariate normal distribution

Moving into higher dimensions, we now consider the Gaussian target distributions $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. We used $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$ as the initial particle distribution and ran the algorithms for 10^4 iterations with step size 0.1. Whenever necessary for numerical stability, we used a smaller step size and adjusted the number of iterations accordingly.

We compare the approximation quality of the final set of particles generated using Med-SVGD and Ad-SVGD as the dimensionality of the problem increases. Because the Wasserstein 2-distance

between two Gaussian distributions has an explicit formula (see [StatisticalAspectsWasserstein]), we use as a measure of sample quality the Wasserstein 2-distance between the target distribution π_d and the Gaussian distribution $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical mean and covariance matrix calculated from the set of particles [StatisticalAspectsWasserstein]. In accordance with the corresponding function of the Python Optimal Transport library [PythonOptimalTransport], which we used for our calculations, we call this the *Bures Wasserstein distance*. Figure 10a shows the development of this sample quality measure, achieved using Med-SVGD and Ad-SVGD with $M = 50, 100, 200$, as the dimensionality of the problem increases. We see that Ad-SVGD significantly outperforms Med-SVGD for all values of d .



(a) Approximation error for $M \in \{50, 100, 200\}$.

(b) χ^2 -test statistic for $M = 200$.

Figure 10: Results for multivariate normal distributions of increasing dimension

For the Bures Wasserstein distance, the smaller variances in the last components of our target distributions π_d do not have a big impact; to see the impact of failing to correctly capture the uncertainty in these components, we consider the test statistic $X \cdot \Sigma_d^{-1} X$ which is χ^2 -distributed with d degrees of freedom (i.e. it has expected value d) for $X \sim \mathcal{N}(0, \Sigma_d)$ (cf. [Metropolisadjustedinteracting]). We calculate the mean of this statistic on the set of particles with $M = 200$ and compare it to the expected value d in Figure 10b. Moving beyond the one-dimensional case, the test statistic for Med-SVGD deviates further and further from the true expected value as d increases, while the statistic for Ad-SVGD stays relatively close to the true expectation. This shows the failure of Med-SVGD to correctly approximate higher-dimensional distributions, while Ad-SVGD is able to deal with those examples well.

We provide more details about the final particle distributions for $M = 200$ and different d in Tables 1 and 2. They list, for each d , the marginal variances of the set of particles compared to the marginal variances of the target distribution, which are given in the first row. Table 1 shows the results for Med-SVGD, Table 2 shows the results for Ad-SVGD. We see that Ad-SVGD is able to achieve a good approximation of the target distribution in all components in terms of the marginal variances. The particles generated using Med-SVGD, on the other hand, significantly underestimate the uncertainty of the target distribution. This matches the observations already made in Figure 10b.

Lastly, we focus on the marginal particle distributions for $M = 200$ and $d = 8$. To ease the visualization, we normalized them by scaling each component of the particles with the inverse of the corresponding marginal standard deviation of the target distribution (i.e. we multiplied the i -th component with i). This turns each marginal distribution of π_d into a standard normal distribution.

Table 1: Marginal variances of final particle distribution for d -dimensional examples generated using Med-SVGD with $M = 200$; marginal variances of the target distribution π_d shown for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of final particle distribution for d -dimensional examples generated using Ad-SVGD with $M = 200$; marginal variances of the target distribution π_d shown for comparison.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

Figures 11 and 12 show the histograms of these normalized marginal particles distributions for $d = 8$ generated using Med-SVGD and Ad-SVGD, respectively. A standard Gaussian density is shown in each plot for comparison. Again, we observe that Ad-SVGD is able to capture all marginal distributions well, while Med-SVGD underestimates the uncertainty of the target distribution. These observations are also visible in Figures 13 and 14, where the quantiles of the normalized marginal particle distributions are compared against the target quantiles (i.e. against a standard normal distribution).

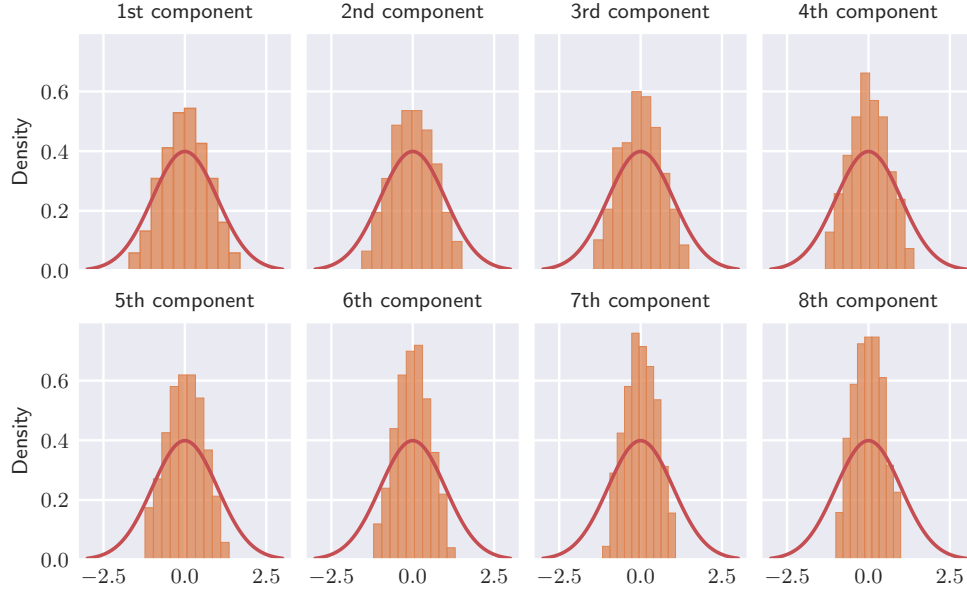


Figure 11: Histograms of single components of the final set of particles for eight-dimensional example generated using Med-SVGD with $M = 200$ and rescaled using marginal target variances; standard Gaussian density shown for comparison.

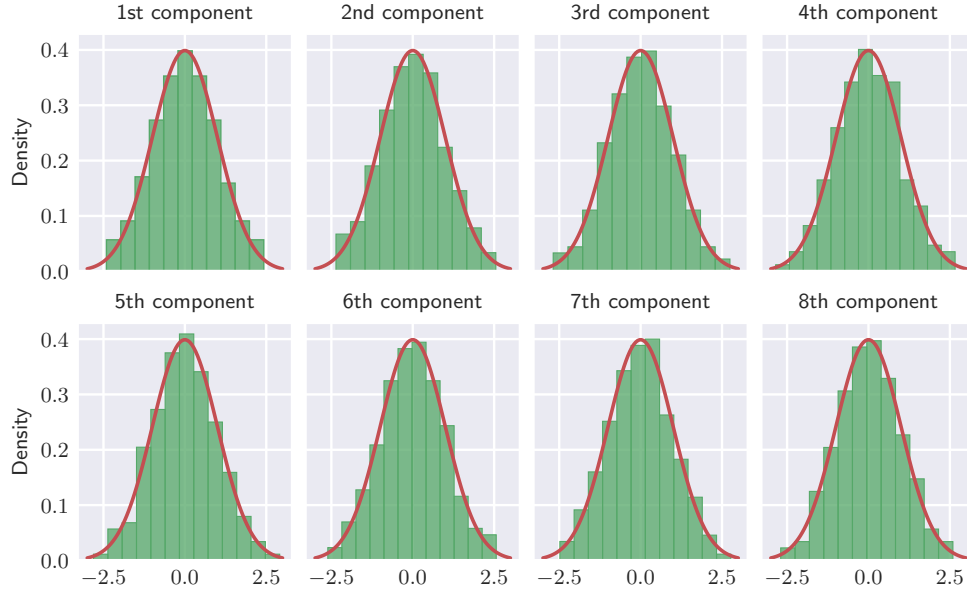


Figure 12: Histograms of single components of the final set of particles for eight-dimensional example generated using Ad-SVGD with $M = 200$ and rescaled using marginal target variances; standard Gaussian density shown for comparison.

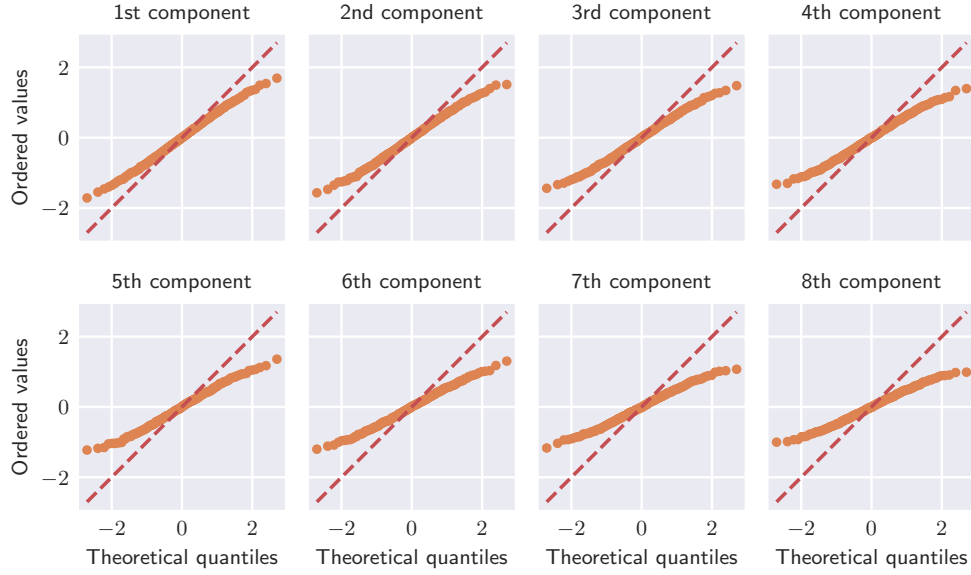


Figure 13: Q-Q plots comparing the marginals of the final particle distribution for eight-dimensional example generated using Med-SVGd with $M = 200$ and rescaled using marginal target variances with a standard normal distribution; line of slope 1 passing through the origin shown for comparison.

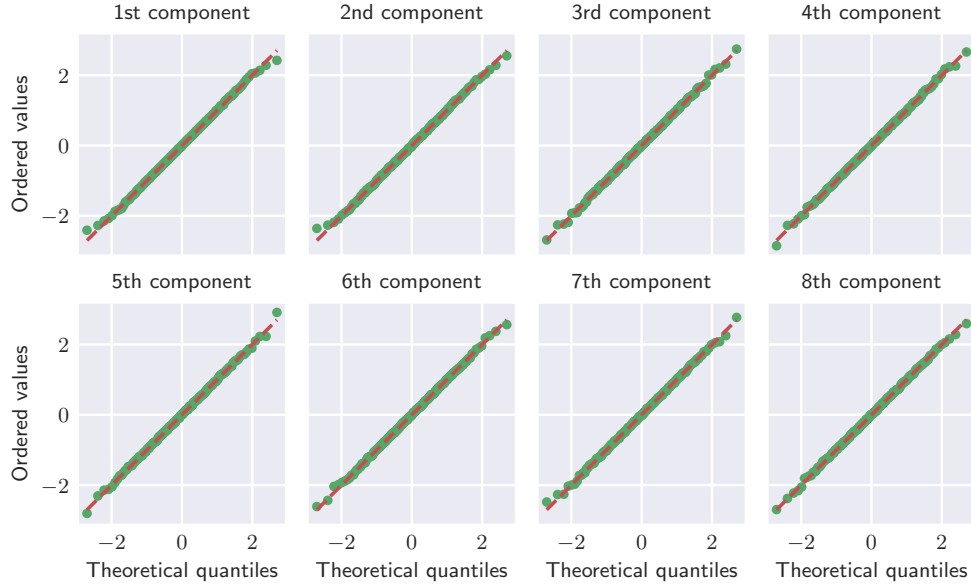


Figure 14: Q-Q plots comparing the marginals of the final particle distribution for eight-dimensional example generated using Ad-SVGd with $M = 200$ and rescaled using marginal target variances with a standard normal distribution; line of slope 1 passing through the origin shown for comparison.

A.3 Gaussian process inference

We demonstrate the behavior of our Ad-SVGD for scaling dimensions in an inference task of a Gaussian process (GP) proposed in [reich21]. We consider a GP on $[0, 1]$ represented by a truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ is a vector of independent Gaussian random variables $x_k \sim \mathcal{N}(0, k^{-2})$. We observe the process at N_y equispaced points in $[0, 1]$ and infer the coefficients x_k . For fixed N_x and N_y , this corresponds to an inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$, $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal matrix Σ with entries k^{-2} , $k = 1, \dots, N_x$ and we assume independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. We construct reference observations $\bar{y} \in \mathbb{R}^{N_y}$ by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

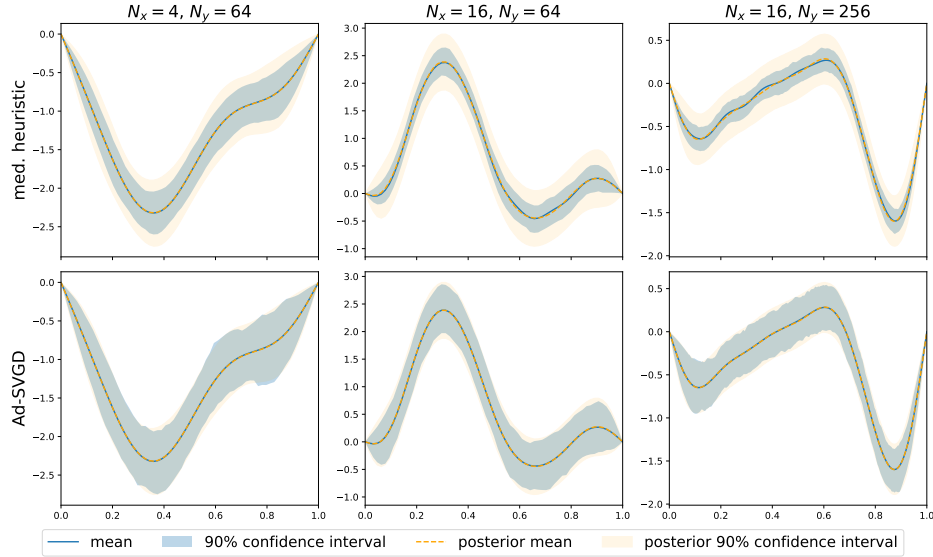


Figure 15: Estimated processes generated using the median heuristic and Ad-SVGD with $M = 100$ particles for different model configurations compared to posterior.

We use SVGD to sample from the resulting posterior $X | Y = \bar{y}$ and compare the performance of the median heuristic with our adaptive approach. For SVGD with the median heuristic, we use kernels of the form $k(x, y) = \exp(-\|x - y\|_1/h)$; for Ad-SVGD, we use product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with parameter $h = (h_1, \dots, h_{N_x})$, i.e. we use a different bandwidth for each dimension. Following [SVGD], we use a variant of Adagrad for step size control in the scenarios with $N_x = 16$.

Figure 15 shows the resulting processes in comparison to the posterior for the choices $N_x = 4$ and $N_y = 64$, $N_x = 16$ and $N_y = 64$ as well as $N_x = 16$ and $N_y = 256$. As the dimension increases, SVGD with the median heuristic underestimates the posterior variance, while Ad-SVGD is able to give a better approximation. The behavior is consistent across different numbers of observations of the Gaussian process. In Table 3, we quantify these results by comparing the trace of the covariance of the particle distributions generated by SVGD with the true posterior. SVGD with the median heuristic severely underestimates the uncertainty while Ad-SVGD is able to capture the variance more accurately.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of covariance of final particle distribution ($M = 100$ particles) compared to theoretical posterior for different configurations, averaged over 25 runs.

B Refined theoretical analysis

We can further improve our convergence result under a so-called Stein log-Sobolev inequality, which relates the KL divergence to the KSD. **Duncan2019OnTG** introduced this inequality and discussed conditions under which it might hold. Interestingly, in our adaptive setting of kernel selection, we can relax this condition by requiring only one kernel in our entire kernel class to satisfy the log-Sobolev inequality. More precisely, we make the following assumption.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \mid \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Using this property, we can quantify convergence in terms of the KL divergence. We emphasize that in this scenario we can give an error bound on $\mathcal{F}(\mu_n)$ even under an approximate condition $\varepsilon_n \leq \bar{\varepsilon}$ without requiring $\varepsilon_n \rightarrow 0$.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\|) + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) - \text{KSD}_{\theta_n}^2(\mu_n \mid \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we deduce the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We consider the three different cases separately:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows by the computation of the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

where we have used the formula for geometric series. Similarly, in the case of $q < \rho$, we can directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, in the case $q = \rho$, we simply bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and deduce the claim line by line as before.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here, we have used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□